

excluding the requirement for suitable land conditions. Ground level is used for fishing and storage. Benefits include no dependence on rainfall, reduced imports and faster time-to-market.

Many a time, even after sufficient cultivation, food wastage at one place results in deficiency in another. One of the main reasons behind this is absence of cold storage, as there is a lack of access to electricity in many underdeveloped areas in developing countries. Solar refrigerators, like ColdHubs, provide a sustainable way of storing perishable food items and can reduce wastage significantly by increasing their shelf life through heavy insulation. Thermal energy

from sunlight is sufficient to convert the refrigerant from gaseous form to liquid, which returns to gaseous form once heat is removed. Operational cost gets reduced due to higher energy efficiency and absence of electrical source or DC to AC converter.

We all know how plants make their food through photosynthesis. Now, scientists have found a way to use carbon-dioxide and sunlight in a similar manner. In 2017, Lappeenranta University of Technology and VTT Technical Research Centre of Finland jointly produced a protein using carbon-dioxide and electricity from a clean source of energy for fermentation. This protein can be fed to cattle as well.

Finland-based Solar Foods plans to bring the protein powder prepared by extracting carbon-dioxide from air using carbon-capture technology (like that provided by Global Thermostat), water, nutrients and vitamins under the name Solein into the market by 2021. As per reports, with fifty per cent protein, 20 to 25 per cent carbohydrates and five to ten per cent fats, this could be an ingredient in a wide variety of applications.

Although such carbon-neutral initiatives are increasing, without effective implementation, the world target of reaching zero hunger by 2030 will not be possible.

—Ayushee Sharma

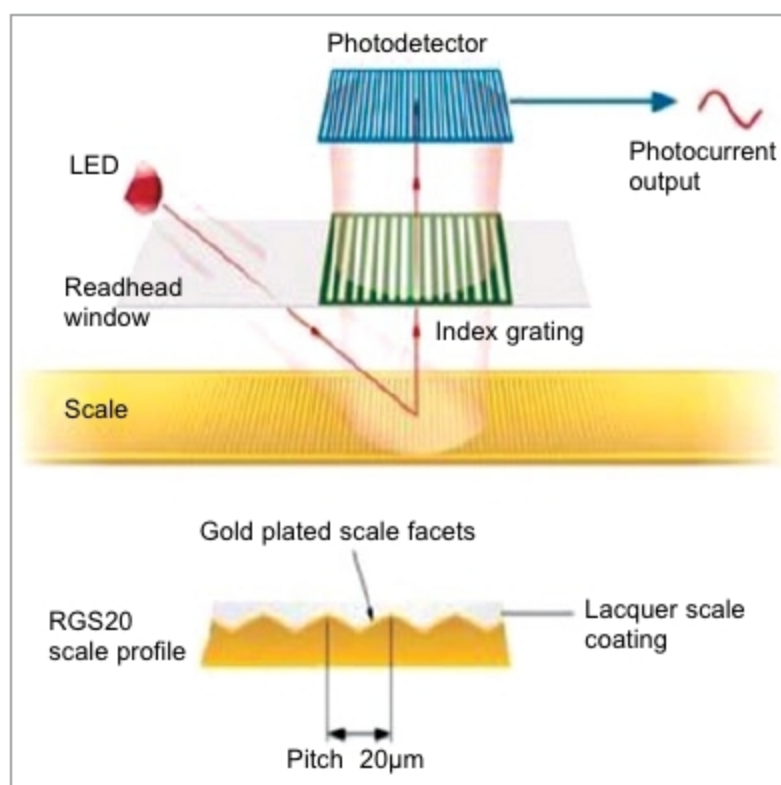
2 BRINGING ADVANCED PRECISION TO LASER MARKING SYSTEMS

In addition to saving time, laser technology improves safety and results by utilising miniature optical encoders with high resolution and accuracy to focus exactly on the target spot in surgeries usually done on the human eye, giving it an edge over traditional methods

Latest applications require improvements in design and manufacture using advanced technologies, especially where size and mass are critical. Developments in robotics and Industry 4.0 have made a significant contribution to the growing possibilities in this market. Laser technology is one such example, where remarkable precision is required to ensure effectiveness.

In the laser-marking process, surfaces of different materials including metals (such as aluminium, copper, gold, silver and platinum), plastics (such as polycarbonate and ABS) and paints get labelled using a low-powered laser beam. Errors in beam or sensor positioning results in damage to products that need to be marked or engraved.

To project a laser beam at the required location while laser-marking items, position of the mirrors is steered using linear motion of voice coil motors. China-based Han's Laser Group, a global professional laser equipment manufacturer, required a non-contact, compact and highly-accurate optical encoder for control-



(Credit: www.renishaw.com)

ling velocity and angular position of voice coil motors through feedback.

An optical encoder identifies position change as light passes through a patterned encoder wheel or disk. Selecting the right model and knowing the appropriate configuration is crucial for success. There are several encoder manufacturing companies in

the market such as Broadcom, Yuheng Optics, Ifm electronic, Siko GmbH, Allied Motion, Posital Fraba, etc. Among them, one company is Renishaw whose RGH24 incremental encoder system with RGS20 linear scale was chosen as a reliable solution for high dynamic performance. An incremental encoder generates a pulse per step in rotation to determine position and speed, and reports position changes in real time.

Though optical encoders are inherently analogue, RGH24 is a tiny read head (36mm × 13.5mm ×

14.8mm) that allows both analogue and digital outputs with a capability of integral interpolation up to 10nm for high linear speeds. Sub-divisional error remains as low as ±0.15µm. It has a setup LED that becomes green when installation gets completed accurately and filtering optics for dirt immunity.

RGS20-S tape scale is supplied on